Pixels, Frames & Transformers

Building an Image Corpus · Module 02

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Collection Comes First

Images are easy to find and hard to use properly. Four questions decide what you can do with a corpus once you have it, so answer them **before** the first download:

Rights
Who holds the copyright, and under which license can I use the images?

Platforms
What do the terms of the site or archive allow: automated access, research use?

People
Do the images show identifiable people? What is my legal basis, and how do I protect them?

Sharing
What will I need to publish (figures, data, a replication package), and am I allowed to?

Where Research Images Come From

Source	What you get	Typical constraints
Archives and museums	curated collections with rich metadata, increasingly openly licensed	reuse terms vary per collection
Wikimedia Commons	freely licensed images, each with author and license	categories are volunteer-made and messy
Open data projects	e.g. Open Food Facts, government photo collections	usually open licenses with attribution
Platform research access	YouTube Data API, TikTok Research API, Meta Content Library, the EU Digital Services Act (Art. 40)	approval needed; mostly **metadata**, not the media files
Data donation	participants share their own data with you	their consent doesn't cover others in their photos
Scraping websites	whatever is publicly visible	copyright, terms of service and privacy all apply

This module uses two sources that are open by design. The habits carry over to the harder ones, where they matter even more.

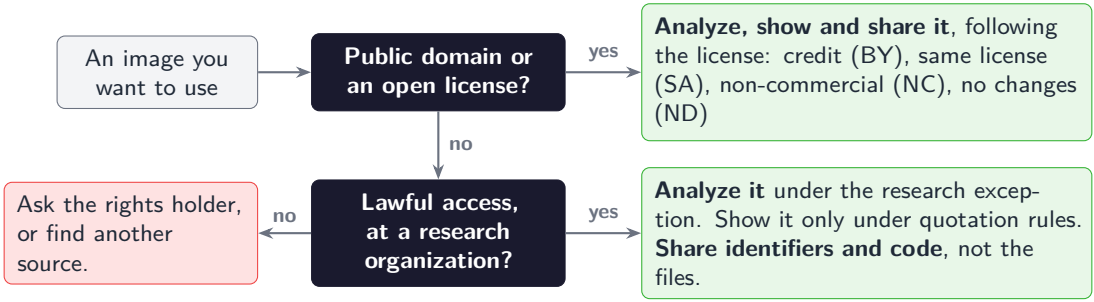
Licenses in Practice

License	Analyze	Show in a paper	Share files	Conditions
Public domain, CC0	✓	✓	✓	none (credit the source anyway)
CC BY	✓	✓	✓	credit the creator, link the license, note changes
CC BY-SA	✓	✓	✓	as BY; anything built from it keeps the license
CC BY-NC	✓	?	✓	non-commercial only; check your publisher
CC BY-ND	✓	✓	✓	no adaptations (cropping can count as one)
All rights reserved	?	?	✗	research exceptions; quotation rules

EU research exception. Article 3 of the Copyright in the Digital Single Market Directive (2019/790) lets research organizations mine works they can lawfully access, images included. That covers *analyzing* them, not redistributing them.

A starting point, not legal advice. Your research data office can help with specific cases.

Can I Use This Image in My Research?



Whatever the license: if the image shows identifiable people, it is **personal data** under the GDPR. You need a legal basis, should collect only what you need, store it securely, and think twice about what you publish.

EU perspective; rules differ elsewhere. A starting point, not legal advice: ask your data steward or privacy officer.

People in Pictures

Faces are personal data
A photo of an identifiable person is personal data, even if you don't know their name. Research is allowed, with safeguards.

Recognition is biometric
Using a face to *identify* someone makes it biometric data, a special category with stricter rules. Analyzing a photo without identifying anyone is not.

Public role or bystander?
A politician at work is not a protester in a crowd, who may not want their participation known.

Minors and what you publish
Children need extra protection. Figures reach far more people than your analysis does: blurring faces is often wise.

Talk to your privacy officer and ethics committee before you collect images of people.

Download Politely, Keep a Manifest

Polite collection

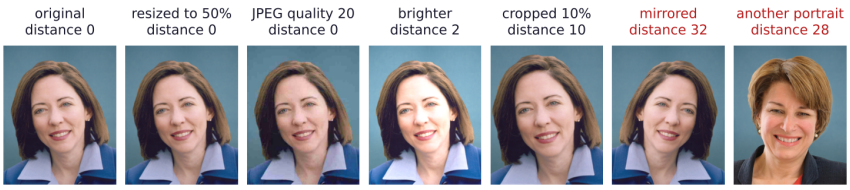
- **Identify yourself** with a User-Agent: project name and contact
- **Pause** between requests and respect stated limits
- **Never download twice**: skip what you have
- **Record failures** instead of crashing
- Thumbnails are often enough

A manifest: one row per image

`local_path`	where the file is on your disk
`source`, `page_url`	where it came from
`license`, `author`	what you may do with it, and whom to credit
`downloaded_at`	when you got it
`sha256`	a checksum: proof the file hasn't changed

Every analysis starts by reading the manifest, not by listing a folder. It is the single most useful thing you can keep about a corpus.

Exact and Near Duplicates



Official portraits of Sen. Maria Cantwell and Sen. Amy Klobuchar, public domain, via unitedstates/images.

Checksum (SHA-256)
changes completely if one bit changes, so it finds only **identical files**. None of these variants shares the original's.

Perceptual hash
64 bits that summarize a tiny grayscale copy. Distance = bits that differ (0–64). Around **10** catches resized and recompressed copies.

Duplicates inflate counts and can land in both training and test data, making a classifier look better than it is.

Share What You Can, Document Everything

You can almost always share

- the **manifest**: identifiers, URLs, licenses, authors, dates, checksums
- your **code** for collecting, filtering and deduplicating
- **derived data**: features, embeddings, codes and labels

With these, others can rebuild your corpus even when the images themselves can't be redistributed.

A datasheet (Gebru et al., 2021)

Motivation: why was the corpus made?
Composition: what is in it, from when and where?
Collection: how were images selected, and what was left out?
Gaps and biases: who and what is missing?
Uses: what should it *not* be used for?

Half an hour of writing that saves your future self, and your reviewers, much more.

In the Notebook

- Four scenarios: licenses, people and publishing
- The Wikimedia Commons API: protest photos with license and author
- Polite downloading and a manifest with checksums
- Breakfast cereal packaging from Open Food Facts
- Exact duplicates with checksums, near duplicates with perceptual hashes
- A datasheet for your corpus

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02_building_an_image_corpus_exercises.ipynb · 7 exercises · CPU runtime · about 10 minutes of downloads